JURONG JUNIOR COLLEGE
JC2 Preliminary Examination 2018

CANDIDATE NAME			
CLASS		INDEX NUMBER	

BIOLOGY
9744/02

Paper 2 Structured Questions

23 August 2018

2 hours

Candidates answer on the Question Paper.

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your class, index number and name in the spaces at the top of this page.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For Examiner's Use	
1	
2	
3	
4	
5	
6	
7	
8	
9	
10	
11	
Total	

This document consists of **27** printed pages and **1** blank page.

Answer **all** the questions in this section.

1 Fig. 1.1 shows an electron micrograph of mitochondria cross-sections.

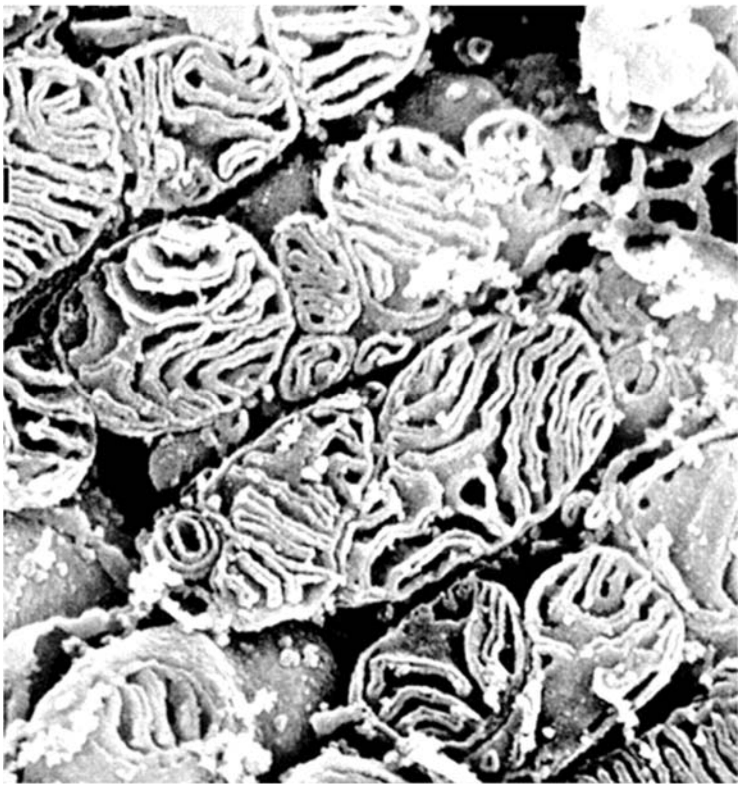


Fig. 1.1

(a) With reference to Fig. 1.1, state one visible feature of the mitochondrion and explain how this feature is adapted for the mitochondrion's function. [3]

Fig. 1.2 shows the molecular structure of carbonyl cyanide-4-(trifluoromethoxy) phenylhydrazone (FCCP). FCCP is a respiratory poison that binds protons and transports them across the phospholipid bilayer of the inner mitochondrial membrane. Protons alone are unable to diffuse freely in this manner.

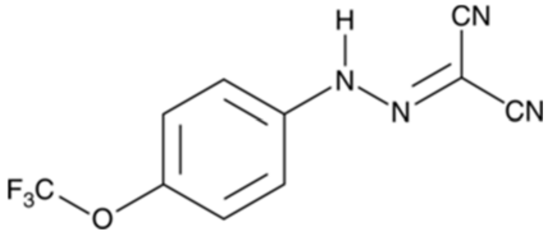


Fig. 1.2

(b) Explain, in relation to their properties, why FCCP readily diffuses across the phospholipid bilayer while protons do not. [2]

____

____

____

____

In the presence of FCCP, the rate of ATP synthesis during respiration is significantly diminished. The rate is further reduced if oxygen becomes unavailable. In such conditions where oxygen is depleted, cells continue to oxidise glucose to form pyruvate and sustain ATP synthesis.

(c) (i) Identify the key stages of aerobic respiration where the release of carbon dioxide occurs. [1]

____

____

(ii) State the location where the biochemical pathway enabling the continual oxidation of glucose to form pyruvate occurs. [1]

____

(iii) Explain how ATP synthesis can be sustained in the absence of oxygen in yeast cells. [3]

[Total: 10]

2 Fig. 2.1 shows a triglyceride molecule.

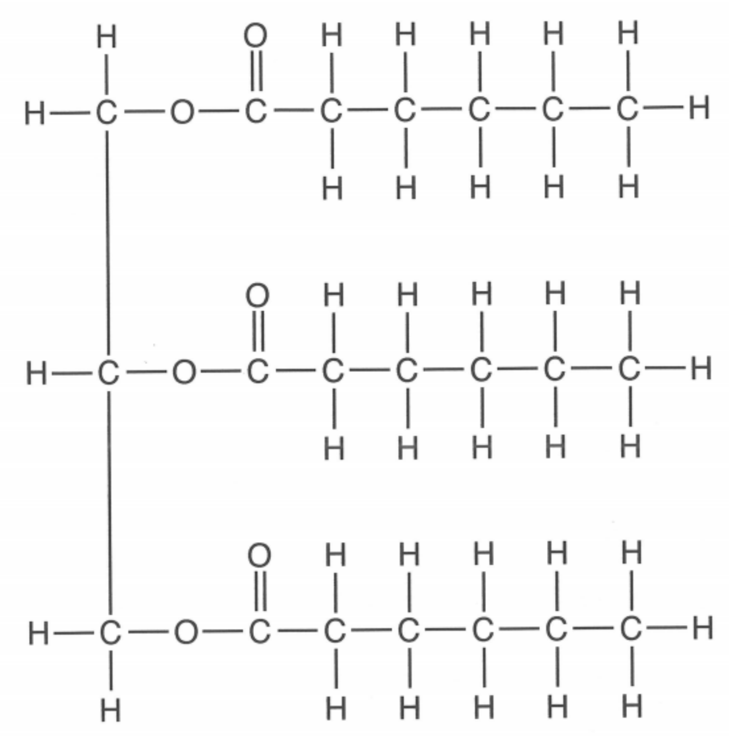


Fig. 2.1

(a) (i) State the names of the two types of molecules that undergo condensation reactions to form a triglyceride. [2]

(ii) Describe what is meant by a condensation reaction. [2]

(iii) The triglyceride in Fig. 2.1 is saturated.

Explain how the structure would be different for an unsaturated triglyceride. [3]

(b) The eukaryotic cell surface membrane contains phospholipids, cholesterol and proteins.

(i) Describe how a phospholipid molecule differs from a triglyceride molecule. [2]

(ii) Describe the roles of cholesterol in eukaryotic cell surface membranes. [2]

(c) The respiratory quotient, RQ, is used to show which substrate is being metabolised by cells. It can be determined using the equation below.

$$RQ = \frac{\text{molecules of carbon dioxide released}}{\text{molecules of oxygen taken in}}$$

Lauric acid is a saturated fatty acid found in coconuts and has a chain of 12 carbon atoms.

(i) Complete the equation below which outlines the aerobic respiration of lauric acid. [1]

$C_{12}H_{24}O_2$ + ……. O_2 → $12CO_2$ + $12H_2O$

(ii) Calculate the RQ value for lauric acid.

Give your answer to 2 decimal places.

RQ value = …………. [1]

[Total: 13]

3 Lysosomes are important membrane-bound organelles within animal cells.

They contain many different types of enzymes that hydrolyse lipids, nucleic acids, polysaccharides or proteins. When phagocytic vesicles containing these food substances fuse with lysosomes, the internal pH of lysosomes will be lowered to a range of pH 4.5 to pH 5.0 resulting in the activation of the lysosome enzymes.

(a) Describe how proteins are hydrolysed.

You may use annotated diagrams to illustrate your answer. [4]

(b) Explain how the lysosome enzymes can be activated by the change in pH. [3]

(c) Suggest why the lysosome membrane is not destroyed by the enzymes in the lysosome. [3]

[Total: 10]

4 Fig. 4.1 shows the elongation phase of translation.

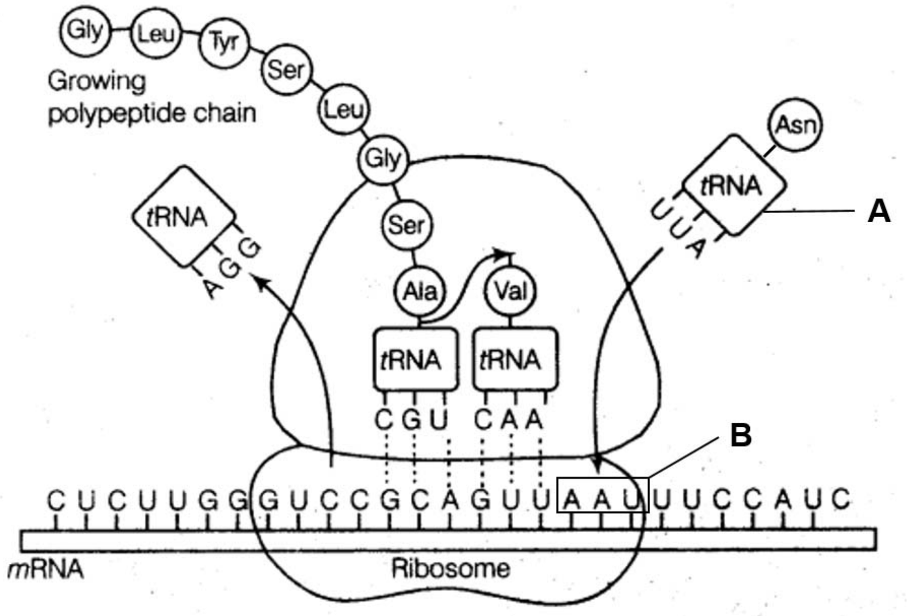


Fig. 4.1

(a) On Fig. 4.1, draw an arrow to show the direction of translocation of the ribosome. [1]

(b) Name the structures **A** and **B** in Fig. 4.1. [2]

A ____

B ____

(c) Explain how the molecular structure of **A** is related to its functions. [2]

____

____

____

____

(d) Describe the phase of translation that occurs before elongation. [3]

____

____

____

____

____

____

(e) State two differences between translation in prokaryotes and translation in eukaryotes. [2]

[Total: 10]

5 Viruses such as bacteriophages have been described as being “organisms at the edge of life”.

(a) State one reason why viruses may be classified as:

(i) living [1]

____

____

(ii) non-living. [1]

____

____

(b) Fig. 5.1 is an electron micrograph of a T4 bacteriophage.

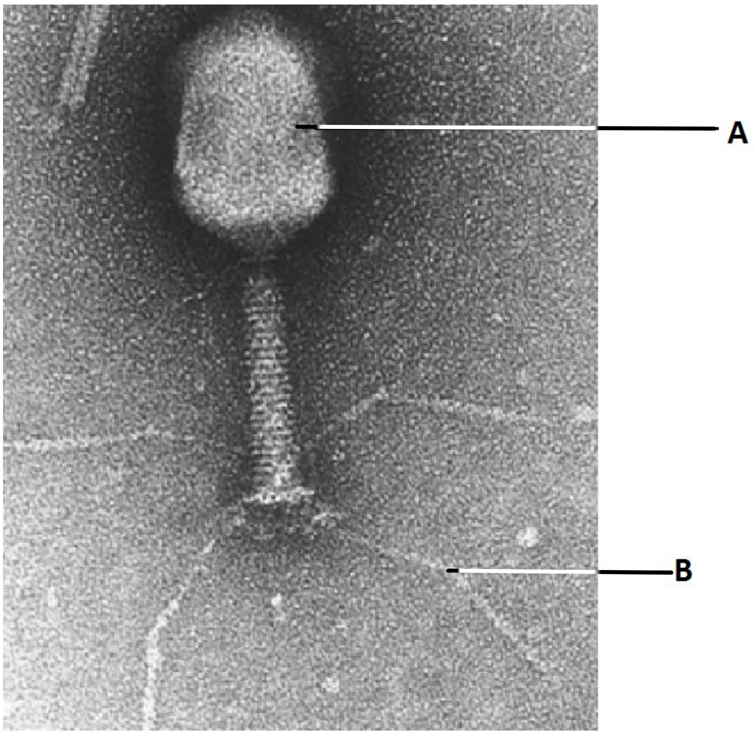


Fig. 5.1

(i) Explain what is meant by the term *bacteriophage*. [1]

____

____

(ii) Identify the structures labelled **A** and **B**. [2]

A ____

B ____

Fig. 5.2 shows a scanning electron micrograph of numerous T4 bacteriophages infecting *Escherichia coli*.

Fig. 5.2

(c) With reference to Fig. 5.2, outline the process which allows for *E.coli* DNA to be transferred to another bacterial cell after infection. [4]

____

____

____

____

____

____

____

____

[Total: 9]

6 In *Escherichia coli*, the level of transcription for most genes varies widely according to the nutrient growth condition. The cause of these differences may be due to the presence of many operon-specific activators or repressors, which vary with the composition of the growth medium.

Fig. 6.1 shows the interaction between a lac repressor, coded by a regulatory gene, and the *lac* operon.

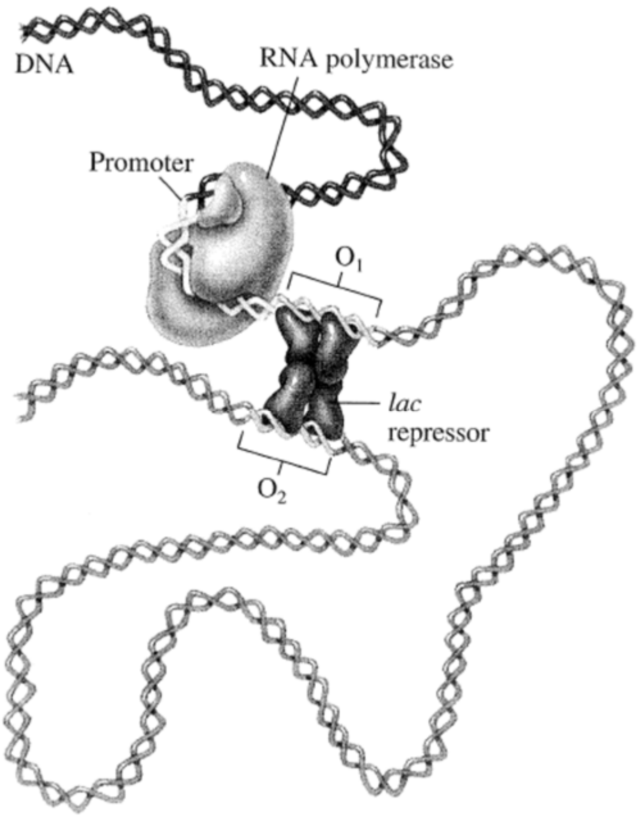


Fig. 6.1

(a) Explain what is meant by *regulatory gene*. [1]

____

____

(b) Name two enzymes coded by structural genes of the *lac* operon. [2]

____

____

(c) With reference to Fig. 6.1,

(i) suggest the nutrient growth condition of the *E. coli* [1]

____

____

(ii) outline how the lac repressor interacts with the *lac* operon. [2]

____

____

____

____

Fig. 6.2 shows the *trp* operon of *E. coli* with five genes *trpA, trpB, trpC, trpD* and *trpE* that code for enzymes responsible for the biosynthesis of the amino acid tryptophan (Trp).

This operon is transcriptionally regulated by trp repressor, encoded by the *trpR* gene, which is located upstream of *trp* operon.

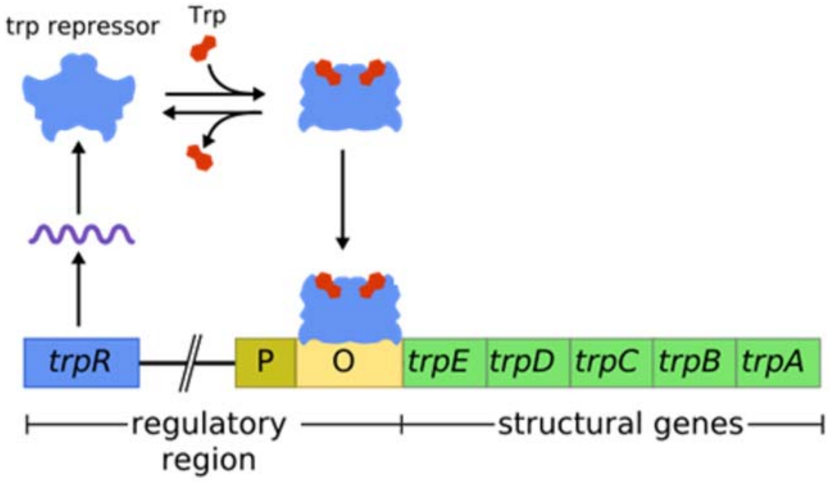


Fig. 6.2

(d) *trp* operon is an example of repressible operon while *lac* operon is an example of inducible operon.

State two other differences between *trp* operon and *lac* operon. [2]

____

____

____

____

[Total: 8]

7 Fig. 7.1 shows an electron micrograph of a chromosome in prophase II.

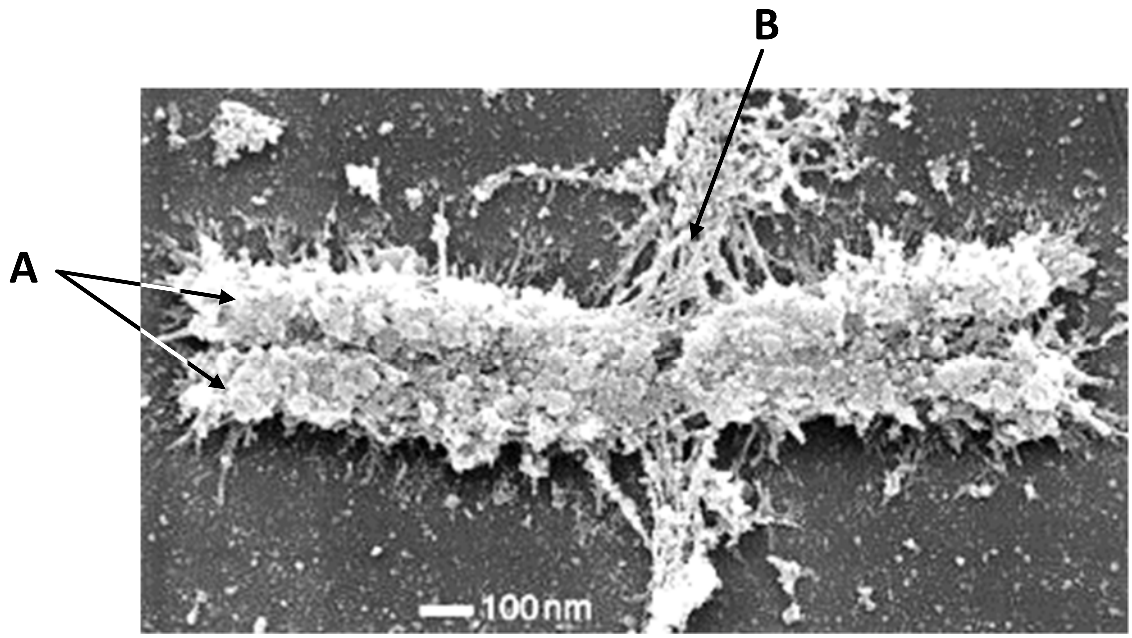


Fig. 7.1

(a) Name the structures **A** and **B**. [2]

A ____

B ____

(b) Explain why the chromosome occurs as a double structure. [2]

____

____

____

____

The risk of mis-segregation of chromosomes increases with age among women. This can lead to aneuploid embryos. Fig. 7.2 shows an oocyte undergoing nuclear division with mis-segregated and lagging chromosomes.

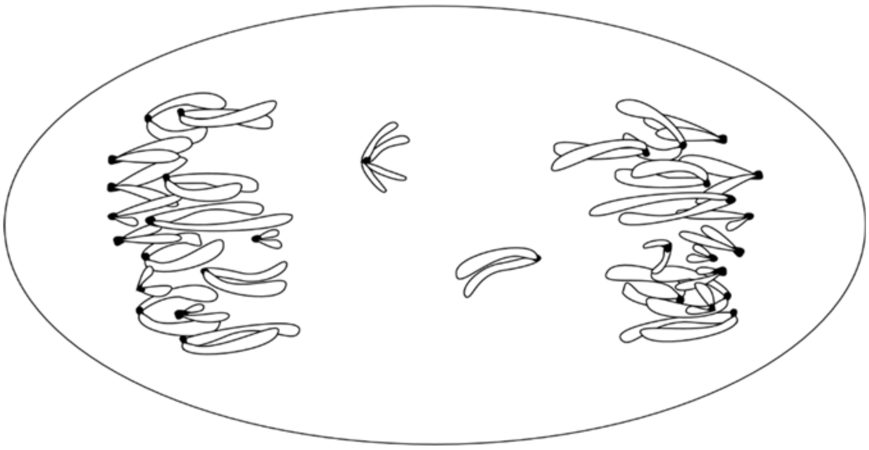


Fig. 7.2

(c) Identify the stage in meiosis as shown in Fig. 7.2. [1]

____

(d) With reference to Fig. 7.2, suggest how aneuploid embryos are formed. [3]

____

____

____

____

____

____

[Total: 8]

8 The sweet pea, *Lathyrus odoratus*, is a flowering plant that grows in many parts of Europe. The inheritance of flower colour and shape of pollen grains is controlled by genes that display autosomal linkage.

(a) Explain what is meant by *autosomal linkage*. [2]

(b) In sweet peas, the allele for purple flowers is dominant to the allele for red flowers and the allele for long pollen grains is dominant to the allele for round pollen grains.

A sweet pea plant that is heterozygous for both purple flowers and long pollen grains is crossed with a sweet pea plant with red flowers and round pollen grains. The results of this cross are shown in Table 8.1.

Table 8.1

offspring phenotype	number of offspring
purple flowers, long pollen grains	44
red flowers, round pollen grains	44
purple flowers, round pollen grains	6
red flowers, long pollen grains	6

Describe and explain the results shown in Table 8.1. [3]

(c) Draw a genetic diagram to explain the results shown in Table 8.1. [3]

Use the following symbols to represent the different alleles involved:

R - purple flower **r** - red flower
L - long pollen grain **l** - round pollen grain

A horticulturist suggested the hypothesis that the phenotypic ratio of the offspring from the cross was 1:1:1:1. A chi-squared test was carried out on the results of the cross.

phenotype	observed number (O)	expected number (E)	$\frac{(O-E)^2}{E}$
purple flowers, long pollen grains	44	25	14.44
red flowers, round pollen grains	44	25	14.44
purple flowers, round pollen grains	6	25	14.44
red flowers, long pollen grains	6	25	14.44
			χ^2 = 57.8

Table 8.2 shows part of the critical values of the chi-squared distribution.

Table 8.2

	Probability								
df	0.995	0.975	0.9	0.5	0.1	0.05	0.025	0.01	0.005
1	.000	.000	0.016	0.455	2.706	3.841	5.024	6.635	7.879
2	0.010	0.051	0.211	1.386	4.605	5.991	7.378	9.210	10.597
3	0.072	0.216	0.584	2.366	6.251	7.815	9.348	11.345	12.838
4	0.207	0.484	1.064	3.357	7.779	9.488	11.143	13.277	14.860
5	0.412	0.831	1.610	4.351	9.236	11.070	12.832	15.086	16.750

(d) Explain the significance of the chi-squared value for these results. [3]

____

____

____

____

____

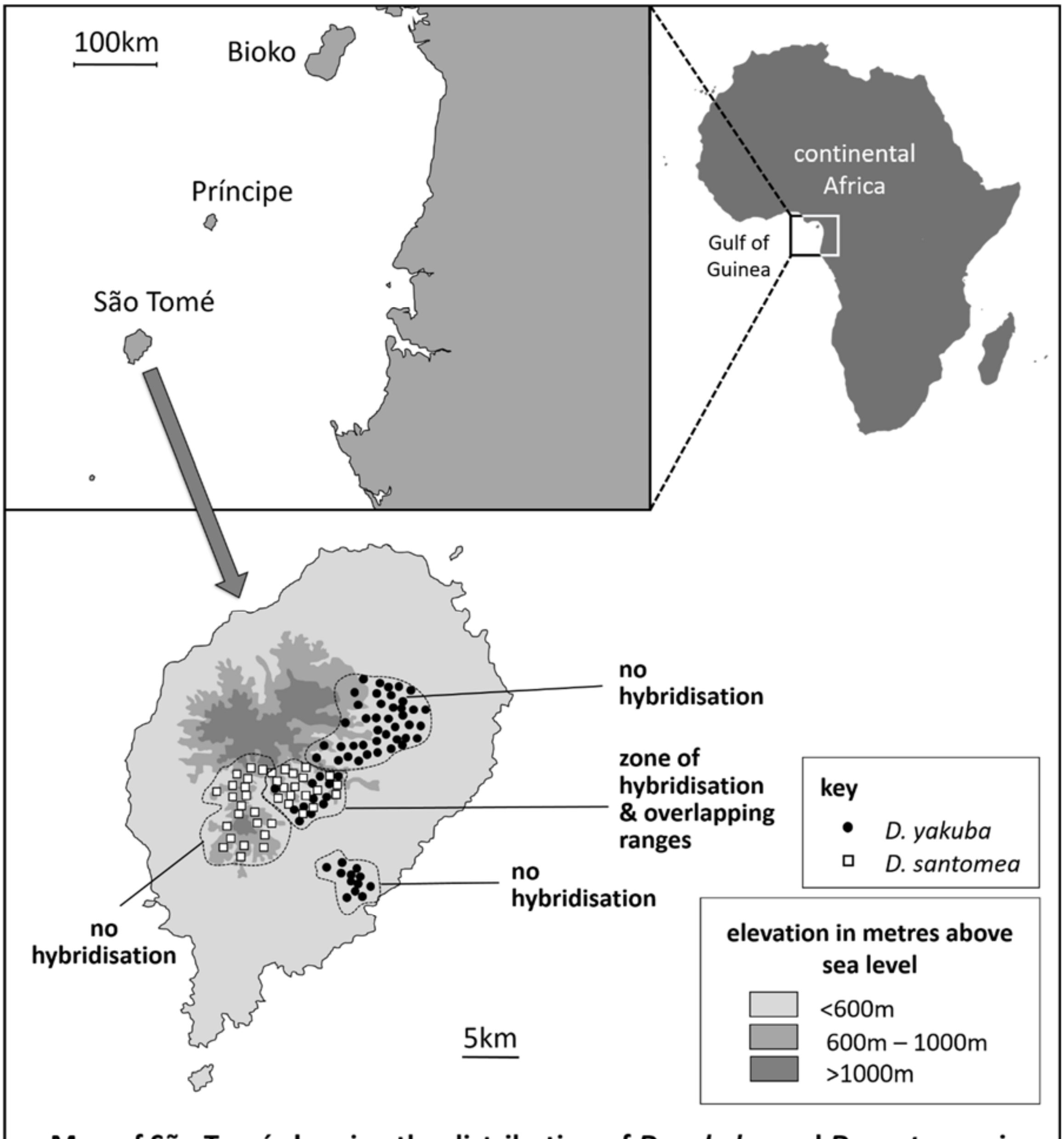
____

[Total: 11]

Question 9 starts on page 22

9 On the volcanic, equatorial island of São Tomé found off the coast of western Africa, two species of vinegar flies *Drosophila yakuba* and *Drosophila santomea* live in co-existence. While *D. santomea* is found only in the moist forests at higher elevations, its close relative, *D. yakuba* resides mostly in the drier lowlands disturbed by human activities. *D. santomea* is also found exclusively on São Tomé while *D. yakuba* can also be found on the neighbouring islands of Príncipe and Bioko and on continental Africa.

Fig. 9.1 shows the distribution patterns of the two species of vinegar flies on São Tomé.



Map of São Tomé showing the distribution of *D. yakuba* and *D. santomea* in relation to elevation above sea level

Fig. 9.1

(a) Give one reason why *D. yakuba* found on São Tomé and continental Africa might be considered as the same species. [1]

(b) With reference to Fig. 9.1, explain how island species like *D. santomea* could have arisen from an ancestral population of vinegar flies. [5]

It was observed that *D. yakuba* and *D. santomea* can hybridise in a well-delineated zone of hybridisation. Hybrids possess intermediate traits of their parents. A population count revealed that hybrids were outnumbered by both *D. yakuba* and *D. santomea* in this zone.

(c) Suggest why the hybrid population is smaller than *D. yakuba* and *D. santomea* populations in this zone. [1]

Studies of the nuclear genome suggested that *D. yakuba* on São Tomé is more closely related to *D. yakuba* from continental Africa than to *D. santomea*. Fig. 9.2 shows the evolutionary relationships among these populations and their sister species *D. teissieri*.

Fig. 9.2

Based on this data, biologists argued that there were two occasions where ancestral populations of vinegar flies arrived on São Tomé from continental Africa and rapidly colonised parts of the island.

(d) Using Fig. 9.2, explain how biologists came to the conclusion that two colonisation events had taken place on São Tomé. [2]

[Total: 9]

10 Phagocytes and lymphocytes are both involved in defence against infectious diseases.

Active B lymphocytes are known as plasma cells.

Fig. 10.1 shows drawings made from electron micrographs of a phagocyte, **A**, and a plasma cell, **B**.

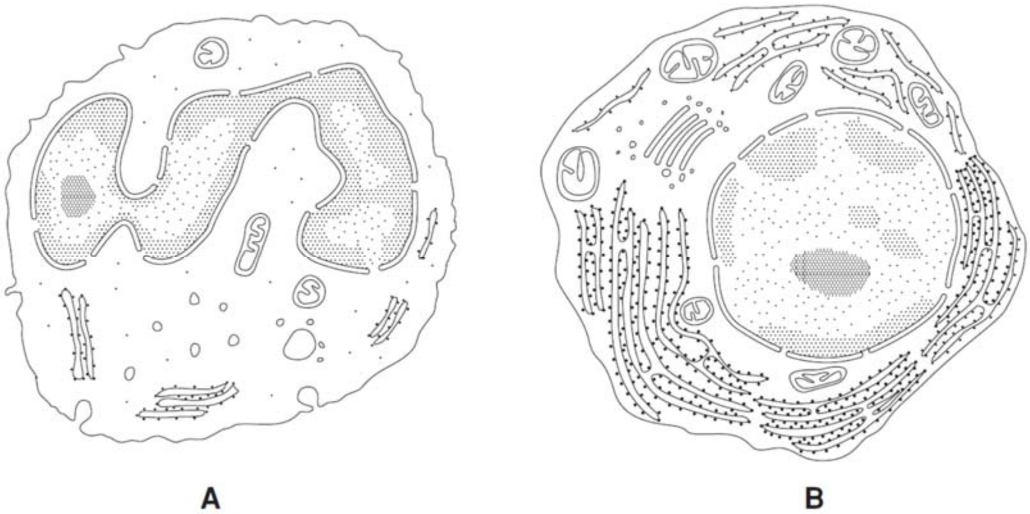


Fig. 10.1

(a) State two visible structural differences between the cells **A** and **B**. [2]

(b) With reference to Fig. 10.1, describe the modes of action of the two cells in defence against infectious diseases. [4]

[Total: 6]

11 An analysis of ice cores from the Arctic and Antarctic can provide information about the composition of the Earth's atmosphere over thousands of years.

Fig. 11.1 shows the concentrations of carbon dioxide measured in ice cores, dated between 1000 and 2000 AD.

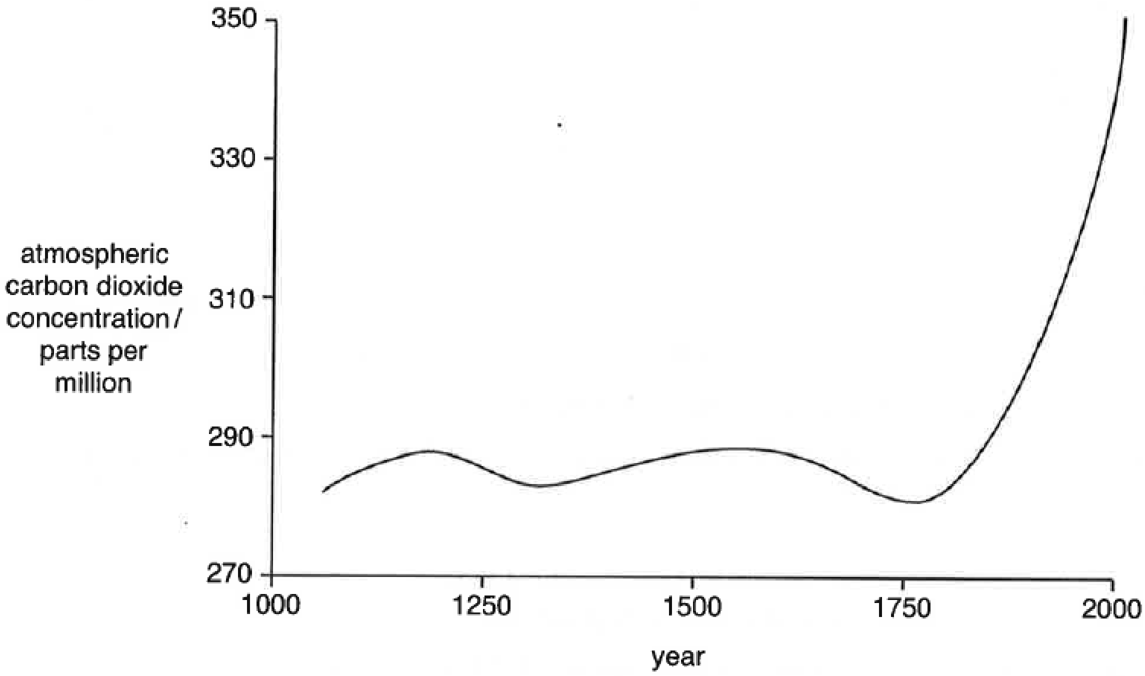


Fig. 11.1

(a) Describe the trend in Fig. 11.1. [2]

(b) Atmospheric carbon dioxide concentrations show regular annual variations. Suggest one reason for this. [1]

(c) Fig. 11.2 shows that, over the same period of time, the average surface temperature of the Earth has shown a similar pattern of change. The increasing concentrations of carbon dioxide is thought to be responsible for the increase in temperature over the last 100 years. This is referred to as the enhanced greenhouse effect.

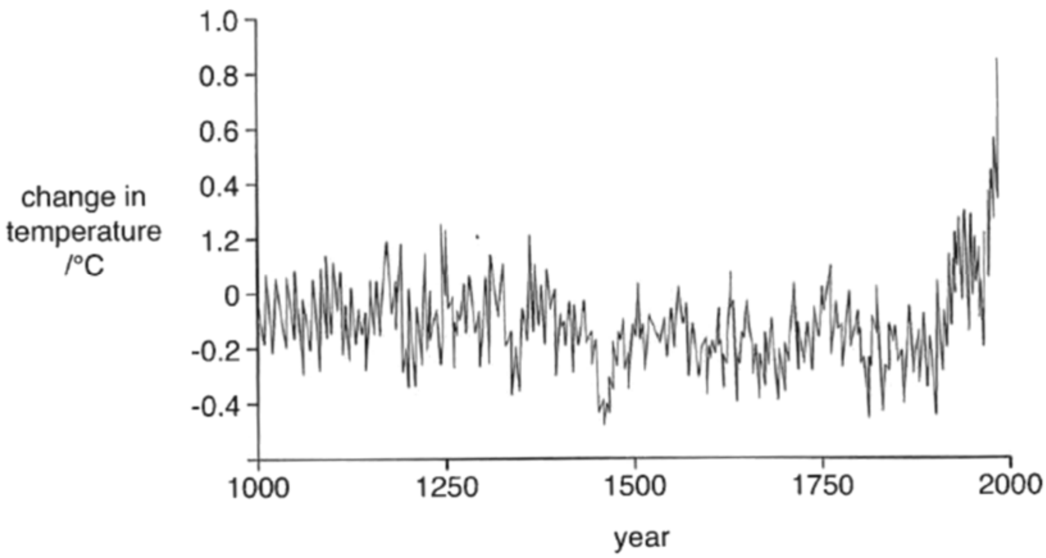


Fig. 11.2

(i) Describe one way in which the data in Fig. 11.2 resembles the data in Fig. 11.1 and one way in which it is different. [2]

Similarity ____

____

Difference ____

____

(ii) Describe one human activity that has contributed to the enhanced greenhouse effect. [1]

____

____

[Total: 6]

BLANK PAGE